

Sushmoy Barua

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Career Objective

Driven by the challenge of "black-box" algorithms in healthcare, my work focuses on developing Explainable AI solutions that prioritize data privacy and security. I specialize in the optimization of Federated Learning protocols to enable secure, collaborative research across medical institutions. I am particularly interested in utilizing Multimodal Fusion to synthesize diverse clinical datasets into interpretable, secure, and robust intelligence for improved patient outcomes.

Research Interest

Deep Learning, Data Privacy & Security, Federated Learning, Healthcare AI, Optimization, Explainable AI

Education

Jahangirnagar University

May 2023 - Oct 2024

Masters of Science in Computer Science & Engineering

- **Supervisor:** Prof. Abu Sayed Md Mostafizur Rahaman
- **Project:** Windows Ransomware Detection: A Federated Learning Approach With TLBO Optimization

Jahangirnagar University

Feb 5 2018 - Jul 2023

Bachelor of Science in Computer Science & Engineering

- **Supervisor:** Assoc. Prof. Sarnali Basak
- **Project:** Diabetic Retinopathy Classification With Transfer Learning And CNN

Technical Skills

- **Programming & Scripting:** C, C++, Python
- **AI & Data Science:** Scikit-learn, NumPy, Pandas, PyTorch, TensorFlow, Image Processing, Federated Learning, Optimization.
- **Development Tools** VS Code, PyCharm, Jupyter Notebook, Google Colab.
- **Web Development Skills:** Javascript, ReactJS, ExpressJS. MongoDB, PHP, MySQL.
- **Soft Skills:** Project management, teamwork, mentoring, technical support, and effective communication.

Undergraduate Research

Windows Ransomware Detection: A Federated Learning Approach With TLBO Optimization

June 2025 – Present

Under Preparation For Journal of Cybersecurity.

- Applied federated learning (FL) to enable decentralized ransomware detection while preserving user privacy.
- Incorporated TLBO optimization to fine-tune the detection model, improving accuracy and adaptability.
- Integrated explainable AI techniques like LIME to enhance model transparency and trust.
- Aimed to provide a scalable and privacy-preserving framework for early ransomware detection and mitigation

Enhancing PCOS Diagnosis: Feature Ranking and Recognition Using MIFS and Explainable AI

Dec 2024 – Present

In Preparation

- Applied MIFS to identify key features for accurate PCOS diagnosis.
- Incorporated explainable AI to enhance model transparency and trust.
- Leveraged machine learning to improve diagnostic accuracy and interpretability.
- Aimed to support healthcare providers in making informed decisions for better PCOS patient management.

Towards Privacy-Aware PCOS Diagnosis: A Federated Learning Approach

Apr 2025 - Present

In Preparation

- Developed a federated learning framework for PCOS classification from medical images.
- Used MobileNetV2 for client-side feature extraction and a centralized neural network as the global model
- Designed a privacy-preserving AI pipeline with three federated clients, improving PCOS detection while ensuring data privacy.

Research Experience

Advanced Intelligent Multidisciplinary Systems (AIMS) Lab

Research Engineer (Data Science)

Dec 2023 – Feb 2024

- Conducted research under the supervision of Prof. Dr. Khondaker Abdullah Al Mamun
- Worked on “AI Tool for Screening Neurodevelopmental Impairments in Children (0-60 Months) in Bangladesh.”
- Contributed to “Smart-NDA: Cloud-Based Framework for Automated Neurodevelopmental Disorder Screening.”

Professional Experience

Samsung Research & Development Institute, Bangladesh

Intern Engineer

Feb 2025 – Feb 2026

- Ensuring product quality through comprehensive testing and validation of advanced software solutions
- Gaining hands-on experience with cutting-edge technologies and large-scale system development
- Exploring multimodal AI and non-invasive cholesterol measurement via smartwatches, aligning with research interests in digital health and biosignal processing

Academic Projects

Bangla Multiclass Cyberbullying Classification Using BanglaBERT & X-AI

Bangla NLP Project

- Designed a model leveraging BanglaBERT to classify Bangla text into six cyberbullying categories.
- Enhanced decision-making transparency using explainable AI techniques.
- Achieved improved classification accuracy with interpretable outputs.

Sign Language Interpreter

GUI-Based Python Project

- Developed a Python-based GUI tool for sign language translation and a companion Sign Dictionary app.
- Followed the SDLC process and collaborated within a 7-member team.
- Documented requirements in LaTeX and used GitHub Actions for Continuous Integration.

Certification & Awards

Complete Web Development - MERN Stack

Programming Hero

Python 101 for Data Science

Cognitiveclass.ai

Introduction to Data Analytics

Bohubrihi.com

SQL and Relational Databases 101

Cognitiveclass.ai

Leadership & Volunteer Experience

- Organized **fundraisers for flood relief**, and tutored **underprivileged children** to promote education and community welfare.
- Served as **Skill Development Secretary**, JU Computer Club (Oct 2019 – Jul 2021): conducted **workshops, webinars, and Arduino sessions** to enhance technical skills among students.
- Managed events as a **Committee Member and Chess Competition Convener**, JU CSE Fest 2019: oversaw logistics, promoted teamwork, and facilitated participant engagement.
- Guided junior researchers in implementing **Machine Learning** models and documenting experimental results for academic submissions
- **Campus Ambassador**, Programming Hero (Sept 2021 – Jun 2022): promoted **MERN stack training, mentoring programs**, and **community learning initiatives**.

References

Abu Sayed Md Mostafizur Rahaman

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